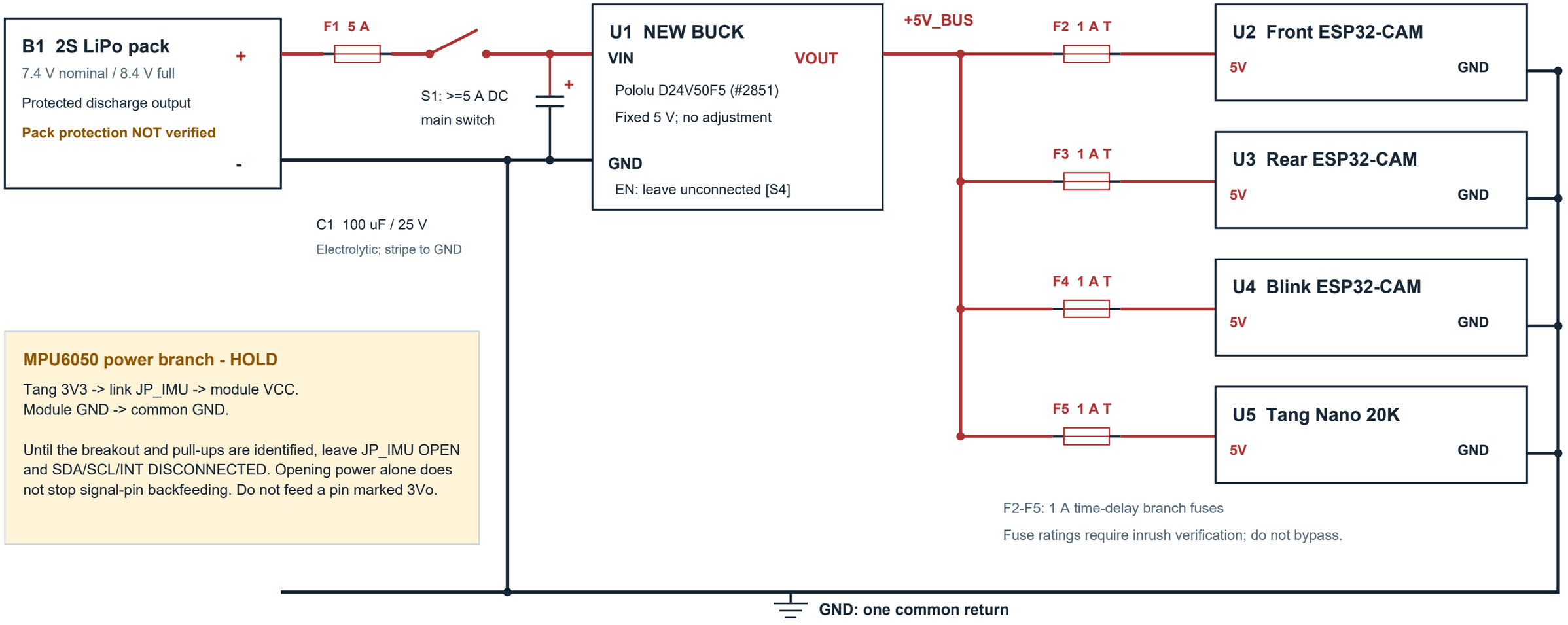
01 Power distribution

One protected 2S LiPo supply, a genuine step-down regulator, and separate fused 5 V branches.

REV C | Replaces earlier wiring tables | MT3608 excluded | Not yet bench-tested



MPU6050 power branch - HOLD

Tang 3V3 -> link JP_IMU -> module VCC.
Module GND -> common GND.

Until the breakout and pull-ups are identified, leave JP_IMU OPEN and SDA/SCL/INT DISCONNECTED. Opening power alone does not stop signal-pin backfeeding. Do not feed a pin marked 3Vo.

Build conditions

Use 20 AWG copper for battery/regulator/trunk wiring and 22 AWG for each board power branch. Use connectors and S1 rated for at least 5 A DC. Keep power runs short; solder or use rated terminals. F1 goes close to the protected battery output.

Supply limits

Design input: 6.4-8.4 V, with cell protection and the pack maker's discharge limit respected. U1 requires about 6.2 V at 5 A; 5 A is thermally limited, not a promise in an enclosure. Allocate 1 A per branch initially; measure actual current and temperature. [S4]

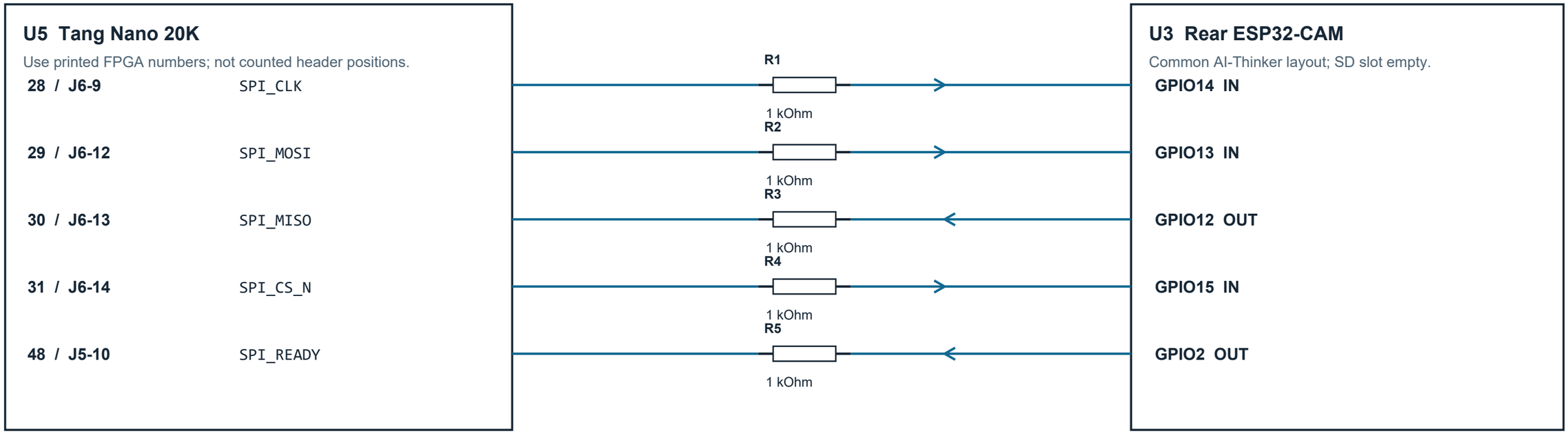
Never connect the MT3608

MT3608 is STEP-UP. Raw LiPo voltage must never reach the 5 V bus. Before USB/programmer power, disconnect that board's power and signal harness. Battery removal alone does not isolate the shared bus. Never join separate 3V3 outputs. The Tang header bypasses its USB fuse. Fuses do not protect against overvoltage. [S1, S3]

02 Rear camera and blink signals

3.3 V logic only. The five rear-camera connections use ESP32 SPI2 plus a ready handshake.

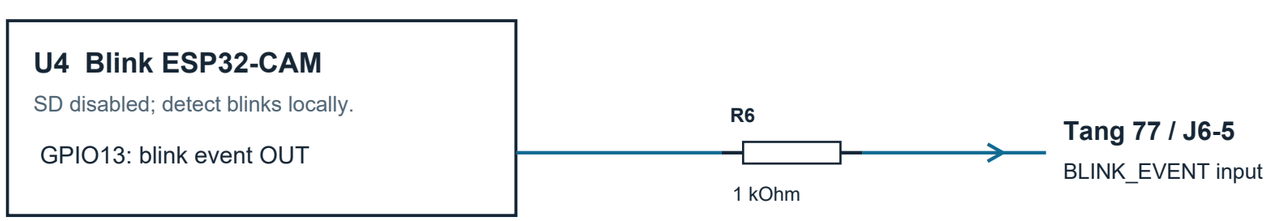
REV C | Replaces earlier wiring tables | MT3608 excluded | Not yet bench-tested



R1-R5: series resistors, one per signal. Start SPI at 1 MHz / mode 1.

SPI protocol and reset requirements

Tang is controller; rear ESP32 is peripheral. Tang drives 28/29/31 and only reads 30/48. Disable FPGA pull-ups on 30/48. Rear READY means a transaction is queued, not just that a photo exists. Send CAPTURE_JEV as a command over MOSI; no separate trigger wire. Use framed chunks with lengths, sequence numbers, and checksums. [S6]



Front ESP32-CAM: power + Wi-Fi only. Each camera remains on its own ribbon connector. No FaceTime hardware in this revision.

Rear ESP32 boot resistors

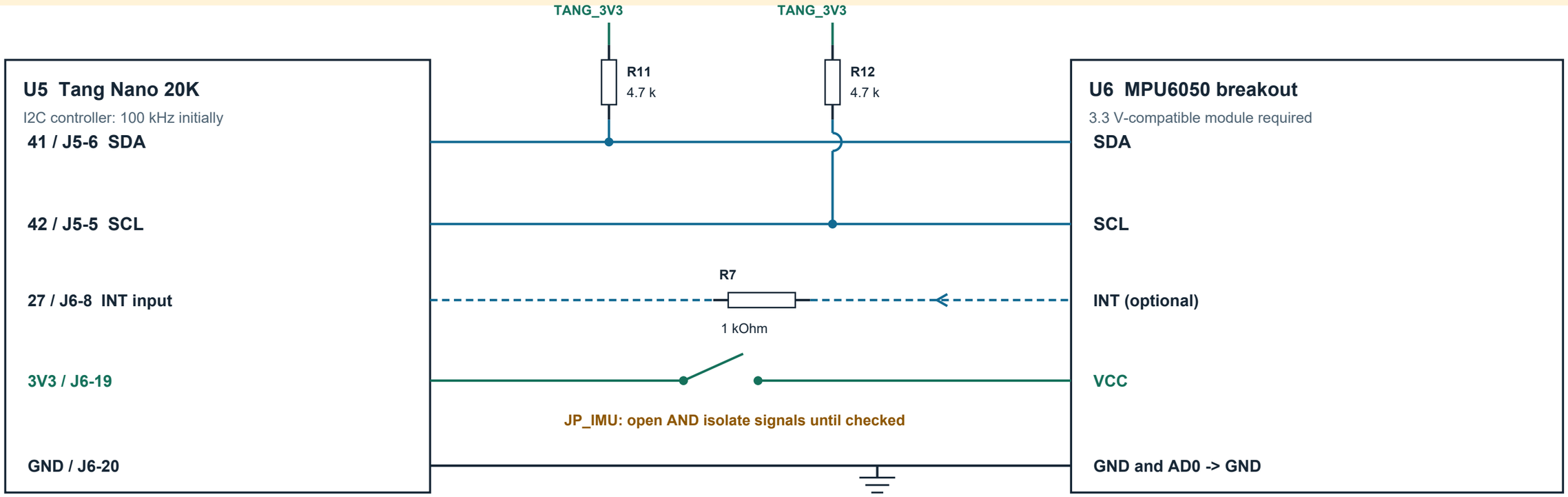


Connect directly at ESP32 pins, before R1-R5. [S5]

03 Motion sensor and speaker

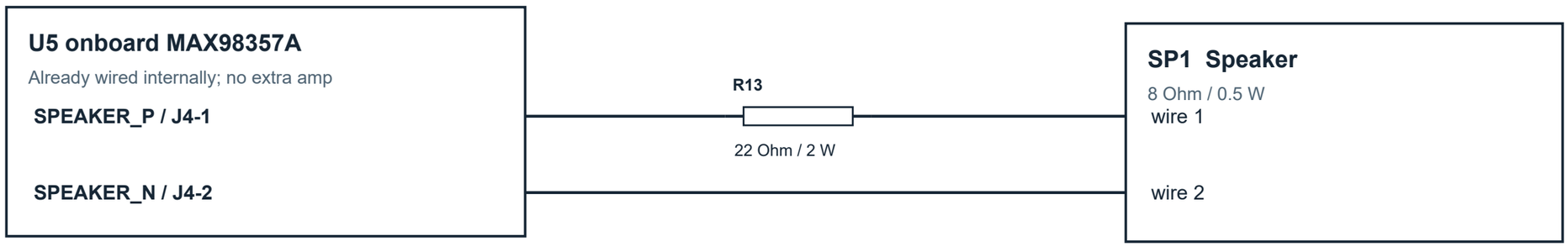
Keep IMU power AND signals disconnected until its breakout is identified. R13 limits speaker power.

REV C | Replaces earlier wiring tables | MT3608 excluded | Not yet bench-tested



I2C details

R11/R12 are NOT fitted if suitable module pull-ups already exist. Pull-ups must use a logic supply compatible with the actual sensor, never 5 V. FPGA SDA/SCL must drive LOW or release (high impedance), never drive HIGH. AD0 low selects 0x68; XDA/XCL stay unconnected. Optional INT can be omitted when polling. [S8]



Neither speaker wire goes to ground.

Keep R13 away from skin and the battery: it can get warm. Leave SP1 disconnected until audio is tested.

Reserve Tang 51=amp enable, 54=data, 55=word clock, 56=bit clock. Start muted and at low PCM amplitude. [S7]

04 Wiring list and parts

Use this revision only. Header references are for the documented Tang Nano 20K; match your board before soldering.

REV C | Replaces earlier wiring tables | MT3608 excluded | Not yet bench-tested

Signal	Tang FPGA / header	Other endpoint	Series part	Tang direction
SPI_CLK	28 / J6-9	Rear ESP32 GPIO14	R1 = 1 kOhm	Output
SPI_MOSI	29 / J6-12	Rear ESP32 GPIO13	R2 = 1 kOhm	Output
SPI_MISO	30 / J6-13	Rear ESP32 GPIO12	R3 = 1 kOhm	Input
SPI_CS_N	31 / J6-14	Rear ESP32 GPIO15	R4 = 1 kOhm	Output
SPI_READY	48 / J5-10	Rear ESP32 GPIO2	R5 = 1 kOhm	Input
IMU_SDA	41 / J5-6	MPU6050 SDA	Direct	Open-drain
IMU_SCL	42 / J5-5	MPU6050 SCL	Direct	Open-drain
IMU_INT	27 / J6-8	MPU6050 INT (optional)	R7 = 1 kOhm	Input
BLINK_EVENT	77 / J6-5	Blink ESP32 GPIO13	R6 = 1 kOhm	Input

Power terminals: Tang 5V=J5-1; GND=J5-2; 3V3=J6-19; GND=J6-20. Never count FPGA numbers as header positions.

Qty	Reference	Part / rating	Notes
1	U1	Pololu D24V50F5, fixed 5 V buck	NEW; MT3608 must not be substituted
1 + 4	F1; F2-F5	5 A main; four 1 A time-delay fuses	DC-rated >=16 V; proper holders
1	S1	SPST switch, >=5 A DC at >=12 V	Disconnects battery positive
1	C1	100 uF / 25 V electrolytic	Positive to switched VIN; negative to GND
6 (+1)	R1-R6 (+R7)	1 kOhm, 1/4 W	R7 optional IMU interrupt
3	R8-R10	10 kOhm, 1/4 W	Rear ESP32 boot-state resistors
0 or 2	R11-R12	4.7 kOhm, 1/4 W	Only if suitable IMU pull-ups are absent
1	R13	22 Ohm, 2 W resistor	Series speaker protection; not 22 kOhm
As needed	Harness	20 AWG trunk; 22 AWG power branches	Insulated connectors; >=5 A main connector

Existing: Tang + 3 ESP32-CAM + 3 OV2640 + MPU6050 + speaker. Still confirm: battery protection, board revisions, IMU breakout.

05 Startup, software and acceptance checks

Wiring is not the firmware. No electrical or functional test has been performed on your physical parts.

REV C | Replaces earlier wiring tables | MT3608 excluded | Not yet bench-tested

Power-up sequence

1. Battery unplugged: verify polarity, insulation and no persistent power-to-ground short. Never use resistance/continuity mode on a powered circuit.
2. Disconnect boards and verify new regulator output in DC VOLTS mode. Target 5.0 V; expected U1 tolerance 4.8-5.2 V.
3. Test each board separately from a current-limited source if available. Inter-board signals and speaker remain disconnected.

Only then connect the system

4. Before USB/programmer power, unplug the target board's 5 V branch and signal harness from the system. Otherwise USB can energize the entire shared bus even with the battery removed. Load the specified directions and LVCMOS33 FPGA I/O.
5. Power off, remove USB, reconnect harness, then test SPI/I2C.
6. Test simultaneous Wi-Fi + FPGA + audio. Check board voltages, peak current, heating and boot reliability. A no-load reading is insufficient.

FPGA responsibilities

Read all three gyro axes over I2C and calibrate stationary bias. Request a fresh Jev snapshot after rotation remains low for a tunable interval (initial trial: 300 ms). Cancel that snapshot if rotation resumes during exposure. Keep collision processing running while turning.

Keep amp enable LOW until clocks and PCM are valid. Synchronize asynchronous READY, blink and INT inputs into the FPGA clock domain.

ESP32 responsibilities

Rear: acquire frames, queue SPI chunks, advertise READY, receive capture commands, upload selected images to Jev and return alerts. Blink: detect locally and emit event pulses. Front: stream/upload to Muse over Wi-Fi.

No microSD firmware on rear/blink boards. Leave camera/PSRAM pins alone. Keep eye-camera flash off. Keep signal wires short (initially <=15 cm), with adjacent ground returns. Muse/Jev API compatibility remains unverified.

Bandwidth and audio checks

160 x 120 x 8-bit grayscale at 10 fps = 192 kB/s. A 1 MHz SPI link carries at most 125 kB/s before overhead: start near 2 fps, then validate 4 MHz or reduce resolution. Wi-Fi/JPEG conversion can interrupt capture; test latency.

R13 limits the ideal 8 Ohm load to about 0.24 W at a 5.2 V full-rail waveform. This is a conservative electrical bound, not a loudness or acoustic safety rating.

Battery and converter requirements

Use the protected discharge terminals of a correctly rated 2S pack. Pack capacity, discharge current, protection and connector polarity remain unverified. Stop before the pack maker's discharge limit; this design uses a 6.4 V operating floor for regulator headroom. U1 UVLO is NOT individual-cell LiPo protection. Use the proper 2S balance charger separately; U1 is not a charger.

Assembly gate - still OPEN

You can build the unpowered harness. Do not connect the LiPo until: U1 has replaced MT3608; your actual Tang/ESP32 layouts match; the IMU breakout is identified; battery protection is confirmed; regulator polarity/voltage are measured; and firmware pin directions are checked. If a fuse opens, find the cause before changing its rating. Do not wear the assembly during first power or thermal tests.

06 Three-pass review and source register

The three checks are design reviews, not three bench tests or a guarantee against assembly errors.

REV C | Replaces earlier wiring tables | MT3608 excluded | Not yet bench-tested

Review	What was checked	Result / remaining work
1 - Power / protection	Reviewed supply topology, current allowance, fuse paths, polarity and speaker power.	MT3608 removed. Fixed buck specified. Header-feed fuse bypass addressed. 22 Ohm / 2 W speaker resistor added. Battery protection and thermal/current tests remain OPEN.
2 - Pins / net conflicts	Traced Tang schematic sheets and compared ESP32 camera, SD and boot pins.	41/42 replace 25/26 for I2C; 77 replaces 49 for blink. SPI pins retained. Audio pins reserved. All FPGA signal assignments unique. Actual board revisions remain OPEN.
3 - Reset / operation	Checked directions, boot pulls, powered/unpowered states, throughput and startup steps.	READY handshake, no GPIO12 pull-up, open-drain I2C, single-source programming, and speaker mute specified. 1 MHz/10 fps mismatch corrected. Physical and firmware validation remain OPEN.

Sources used for this reference design

S1 Sipeed Tang Nano 20K schematic (3921): header rails, display sharing, amplifier and USB fuse

S2 Sipeed Tang Nano 20K datasheet v1.3: header pin labels and supply specification

S3 Aerosemi MT3608 datasheet: step-up topology (excluded from this build)

S4 Pololu D24V50F5 #2851 specifications and manufacturer product documentation

S5 Espressif ESP32 boot-mode selection: GPIO0, GPIO2, GPIO12 and GPIO15

S6 Espressif ESP32 SPI slave driver: SPI2 pins, DMA mode and handshake

S7 Analog Devices MAX98357A/B: bridge output, gain and load characteristics

S8 Joy-IT SEN-MPU6050: example of a documented 3.3 V-compatible breakout; yours is unidentified

S9 Joy-IT ESP32-CAM manual: example common AI-Thinker-style header and shared SD pins

S10 Espressif camera driver: PSRAM/Wi-Fi constraints and pixel-format limitations

Source note: the official schematic text and an earlier 3921 schematic rendering were cross-checked; board revision is therefore a required physical match. This is an original module-level reference design. It is not manufacturer certification, a completed PCB design, or a validated collision-safety system. Click source rows to open documents.